# ANNA Y. CHEN

*Curriculum Vitae*

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**Legal Name:** Yiyu Chen (陈一语)
**Work Authorization:** U.S. Permanent Resident
**Fellowship Applications Submitted:** NSF GRFP
**Fellowship Applications in Preparation:** Soros Fellowship for New Americans; AMS Graduate Fellowship

## Education

2023–2027 **B.A. in Climate System Science & B.A. in Applied Mathematics**, *Columbia College, Columbia University*, New York, USA.

GPA : 4.02/4.00

## Research Experience

### Lamont-Doherty Earth Observatory (LDEO), Columbia University

Spring 2026 – Present **[Senior Thesis] Evaluating the Impacts of Model Choice on Assessment of Surface Ocean $pCO_2$ Reconstruction Skill in Climate Model Testbeds**.

Evaluate 5 Global Ocean Biogeochemical Models and 6 Earth System Models against independent $pCO_2$ data; assess how choosing each as the testbed affects the skill of 7 stats- and ML-based data products. Contribute to the SOCOMv2 initiative.

Mentor : **Dr. Galen McKinley**, *Professor, Dept. of Earth and Environmental Sciences and LDEO*, Columbia U. (Lab web-page)

Spring 2025 – Present **Global Reconstruction of Last Interglacial Sea Surface Temperatures and Their Uncertainties by an Iterated Reduced-Space Optimal Interpolation Procedure**.

Develope gap-filling data product and animation for global sea-surface temperature (SST) using sparse deep-ocean core data over the Last Interglacial. Code in Python. NSF-funded project: HISEAS

Mentor : **Dr. Jerry McManus**, *Professor, Dept. of Earth and Environmental Sciences; Deputy Director for Education, LDEO, Earth and Environmental Sciences*, Columbia U.

Summer 2024 – Spring 2025 **Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity**.

Visualized and analyzed seasonal and geospatial patterns of satellite-derived marine net primary production based on sea-surface temperature and $pCO_2$; coding in R.

Practiced phytoplankton culturing, flow-cytometry, media making, and sterile lab methods.

Mentor : **Dr. Sarah Hurley**, *Assistant Research Professor, LDEO*, Columbia U. (Lab web-page)

## Publications and Manuscripts

**Chen, A. Y.**, McKinley, G. A., Heimdal, T. H., & Fay, A. R. *(in preparation)*. Evaluating the Impacts of Model Choice on Assessment of Surface Ocean $pCO_2$ Reconstruction Skill in Climate Model Testbeds.

Heimdal, T. H., Ford, D. J., Keppens, M., Rogerson, J., Barton, A., Chevallier, F., Roobaert, A., Olivelli, A., Rödenbeck, C., Fay, A. R., McKinley, G. A., Gehlen, M., **Chen., A. Y.**, Shutler, J. D., &

Landschützer, P. *(in preparation)*. The SOCOMv2 initiative: comparison of surface ocean $fCO_2$ reconstructions using the CMIP6-testbed.

Holm, H. C., Schaum, A. P., Ellis, A., Acosta, B., **Chen, A. Y.**, Wing, B., McKinley, G. A., Dyhrman, S. T., & Hurley, S. J. *(in review)*. Using Global Climate Models to Inform Multi-Driver Biological Ocean Experiments. *Submitted to Global Change Biology on Oct 15, 2026.* `https://doi.org/10.31223/X52R6W`.

## Fellowships & Awards

Summer 2026 **Comer Climate Undergraduate Summer Research Fellowship**, *Columbia U.*, ($9,600).
Funds 4 Columbia College students to participate in a 10-week supervised summer research related to climate, sustainability, and environmental justice.

Summer 2025 **Columbia College Summer Research Fellowship**, *Heinrich Family Global Fellowship*, ($3,600).
Awarded to non-seniors at Columbia College for academic excellence to pursue 6-weeks of independent summer research with faculty mentorship

Spring 2025 **Data Science Institute Scholars Program**, *Columbia U.*, ($3,000).
Awarded to mostly Master's, and some highly achieving Bachelor's students, to pursue mentored research in a data-science-related field during an academic semester (15 weeks)

2024-2025 **Laidlaw Undergrad. Research and Leadership Scholarship**, *Columbia U.*, ($7,600 + housing).
Funds Columbia College first-years who wish to pursue summer research, community engagement, and leadership across all disciplines. Full-time stipend awarded across two summers, 6-weeks in each

All Semesters **Dean's List**, *Columbia U.*
Awarded to students with GPA 3.6 and above

2023 **High School Valedictorian**, *YK Pao School*, Shanghai, China.
Received $28,000 in merit-based scholarship. Voted as Class of 2023 graduation ceremony speaker.

## Teaching Assistantship

Spring 2027 **EESC GR4926: Intro to Chemical Oceanography**, Columbia U.

Fall 2025 **EESC UN2100: Earth's Environmental Systems: Climate**, Columbia U.

## Presentations

Fall 2026 **AGU 2026 Annual Meeting**, San Francisco.
1. Oral Presentation: "Evaluating the Impacts of Model Choice on Assessment of Surface Ocean $pCO_2$ Reconstruction Skill in Climate Model Testbeds" *(Session **A24F**)*
2. Poster: "Global Reconstruction of Last Interglacial Sea Surface Temperatures and Their Uncertainties by an Iterated Reduced-Space Optimal Interpolation Procedure" *(Session **PP21C**)*

Fall 2026 **Undergraduate Research Symposium**, Columbia U.
Poster: Evaluating the Impacts of Model Choice on Assessment of Surface Ocean $pCO_2$ Reconstruction Skill in Climate Model Testbeds

Summer 2026 **McKinley/Lovenduski Group Meeting**, U. of Colorado-Boulder.
Talk: "Evaluating the Impacts of Testbed Model Choice to Global Ocean Surface $pCO_2$ Reconstruction"

Spring 2026 **Mini-Symposium on AI-based Science**, LDEO.
Talk: "Can Selecting Testbed (CMIP6) Models Based on the Representation of Seasonality Improve Global Surface-Ocean $pCO_2$ Modeling?"

Fall 2025 **Geochemistry Poster Session**, LDEO.
Poster: "Last Interglacial Sea-Surface Temperature Reconstruction: A Preliminary Approach Using Reduced-Space Optimal Interpolation"

Fall 2025 **North America Laidlaw Scholars Conference**, Brown U.
Talk: "Farming Futures: Building Colorado Communities Through Environmental Stewardship"

Spring 2025 **Scholars Program Poster Presentation**, *Data Science Institute*, Columbia U.
Poster: "Visualizing Sparse Sea Surface Temperature Data From the Last Interglacial"

Fall 2024 **North America Laidlaw Scholars Conference**, Columbia U.
Talk: "Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity"

Fall 2024 **LDEO 75th Anniversary Mini Symposium**, LDEO.
Fall 2024 **Undergraduate Computer and Data Science Research Fair**, *Data Science Institute*, Columbia U.
Fall 2024 **Undergraduate Research Symposium**, Columbia U.
Poster for all 3 above: "Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity"

## Leadership and Volunteering

Spring 2024 – Present **President**, *Zen Buddhism Club*, Columbia U..
Organize speaker-guided meditation weekly and half-day retreats quarter-annually (≈120 ppl. each retreat)

Fall 2024 – Present **President**, *Marie Tharp Society*, Dept. of Earth and Environmental Sciences, Columbia U.
Host social mixers, faculty Q&As, geology movie nights, and student-led Lamont Tours

Fall 2025 – Present **Vice President**, *Columbia U. Cycling Team.*
Coordinate training and racing, apply for school recognition, recruit members, oversee WhatsApp group chat

Fall 2025 – Present **Volunteer**, *Climate Lab*, American Museum of Natural History.
Explain climate processes to museum visitors through hands-on experiments, e.g. the thermohaline circulation

Fall 2024 & 2026 **Volunteer**, *Open House*, LDEO.
Educated children on paleo-botany through making bookmarks from plant-extract paper chromatography

Summer 2024 **Volunteer**, *World Oceans Day*, United Nations.
Facilitated guest check-in and backstage set-up for event with 30+ speakers and 350+ attendees at UN Headquarters

## Languages and Skills

Natural Chinese (native), English (fluent), German (elementary)
Computational Python, R, LaTeX, Markdown
Software Google Suite, MS 365 (Microsoft Suite), iWork (Apple Suite)